



OR 6205: Deterministic Operations Research
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Deterministic Operations Research

Class 16: Duality Theory

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Importance of Weak Duality Theorem

It provides bounds on the optimal objective value z^* . One can find a feasible solution of the dual problem y by inspection. Then, the dual objective value is a valid upper bound on z^* , since $z^* = cx^* \leq yb$.

Example: $z^* = 36$ for the prototype example.

$(y_1, y_2, y_3) = (0.5, 2, 1)$ is a feasible dual solution as it satisfies the dual constraints:

$$y_1 + 3y_3 = 0.5 + 3 = 3.5 > 3; \quad 2y_2 + 2y_3 = 4 + 2 = 6 > 5 \quad \text{and} \quad (y_1, y_2, y_3) \geq (0, 0, 0)$$

This dual feasible solution yields

$$y_0 = 4(.5) + 12(2) + 18(1) = 44 > z^* = 36.$$

Weak duality:

By weak duality, if both the primal and the dual problems have feasible solutions, then:

Weak duality:

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By weak duality, if both the primal and the dual problems have feasible solutions, then both must have an optimal solution.

2. Identify the correct conditions that prove the statement above, where x^0 and y^0 are a primal and a dual feasible point respectively. (Check all that apply.)

☐ $-\infty < cx^0 \leq y^0b < \infty$

☐ $cx \leq y^0b$

☐ $cx^0 \leq yb$

☐ $-1 < cx^0 \leq y^0b < 0$

Weak duality:

→ ☐ $-\infty < cx^0 \leq y^0b < \infty$

→ ☐ $cx \leq y^0b$

→ ☐ $cx^0 \leq yb$

☐ $-1 < cx^0 \leq y^0b < 0$

Let x^0 and y^0 be a primal and a dual feasible point respectively. By weak duality,
 $-\infty < cx^0 \leq y^0b < \infty$

Furthermore, for any primal feasible point x and dual feasible point y ,
 $cx \leq y^0b$ and $cx^0 \leq yb$.

This means that the primal problem cannot be unbounded, as it is bounded above by y^0b and similarly, the dual problem is bounded below by cx^0 . Therefore, both the problems must have an optimal solution.

where x^0 and y^0 are a primal and a dual feasible point respectively.
 (Check all that apply.)

☐ $-\infty < cx^0 \leq y^0b < \infty$

☐ $cx \leq y^0b$

☐ $cx^0 \leq yb$

☐ $-1 < cx^0 \leq y^0b < 0$

LP in Augmented Matrix Form

Duality Theory is based on the Fundamental Insight of LP. Consider a LP in augmented matrix form

Primal problem after slack
vector x_s is added:

$$\begin{aligned} \max z &= cx \\ \text{s.t. } Ax + x_s &= b \\ x, x_s &\geq 0 \end{aligned}$$

Prototype example problem

$$\begin{aligned} \max z &= \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \text{s.t. } \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} &= \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &\geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} &\geq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

Any Simplex Tableau of Primal Using Fundamental Insight Notation

z x x_s RS

(0) Row

1	$yA-c$	y	yb
----------	--------------------------	-----------------------	------------------------

(1 .. m)

0	SA	S	Sb
----------	------------------------	-----------------------	------------------------

Dual of the Prototype Example

In Augmented form (after surplus vector s is subtracted):

$$\min y_0 = \mathbf{y}\mathbf{b}$$

$$\text{s.t. } \mathbf{y}\mathbf{A} - \mathbf{s} = \mathbf{c} \Rightarrow \mathbf{s} = \mathbf{y}\mathbf{A} - \mathbf{c}, \min y_0 = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

$$\mathbf{y}, \mathbf{s} \geq \mathbf{0}$$

$$\text{s.t. } \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} y_4 & y_5 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$$

$$\text{or } \mathbf{s} = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 5 \end{bmatrix} \Rightarrow \begin{bmatrix} y_4 & y_5 \end{bmatrix} = \begin{bmatrix} y_1 + 3y_3 - 3, & 2y_2 + 2y_3 - 5 \end{bmatrix}$$

where th

$$\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \geq \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} y_4 & y_5 \end{bmatrix} \geq \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Observations

The (0) row of the simplex tableau of the primal problem contains a basic solution (BS) of the dual problem!

Why? If we let the dual variable, y_i , to take the value of the coefficient of the i th slack in the (0) row (vector y for all dual variables), then the values of the surplus variables (vectors) are provided by the coefficients of the original (primal) variables (vectors) in the (0) row, according to Fundamental Insight ($s = yA - c$).

Such a dual solution qualifies for a basic solution if it has m 0 values (# of NB variables in the dual problem). It is true as there are m BV in the primal simplex tableau and we expect their coefficients in eq.(0) to be 0.

This dual BS (s, y) , from the (0) row of a primal simplex tableau, is complementary to the primal BS associated with that simplex tableau.

By Fundamental Insight, the primal objective value, $z = yb$, which is the objective function of the dual problem.

So, the respective objective values (z and y_0) of the primal and dual problem for two complementary BSs are equal.

Exercise

Arrange the steps involved in the determination of the primal basic solution to be (0, 10, −10, 0, 0) and the dual basic solution to be (2, 1, 3, 0, 0), in sequential order.

Bas Var	Eq No	Coefficient of						Right side
		Z	X ₁	X ₂	X ₃	X ₄	X ₅	
Z	0	1	3	0	0	2	1	30
X ₃	1	0	3	0	1	−3	2	−10
X ₂	2	0	−1	1	0	2	−1	10

Bas Var	Eq No	Coefficient of						Right side
		Z	X ₁	X ₂	X ₃	X ₄	X ₅	
Z	0	1	−2	−7	−4	0	0	0
X ₄	1	0	1	2	1	1	0	10
X ₅	2	0	3	3*	2	0	1	10

Bas Var	Eq No	Coefficient of						Right side
		Z	X ₁	X ₂	X ₃	X ₄	X ₅	
Z	0	1	5	0	0.667	0	2.333	23.33
X ₄	1	0	−1	0	−0.33*	1	−0.67	3.333
X ₂	2	0	1	1	0.667	0	0.333	3.333

Exercise

Arrange the steps involved in the determination of the primal basic solution to be (0, 10, -10, 0, 0) and the dual basic solution to be (2, 1, 3, 0, 0), in sequential order.

3

Bas Var	Eq No	Coefficient of						Right side
		Z	X_1	X_2	X_3	X_4	X_5	
Z	0	1	3	0	0	2	1	30
X_3	1	0	3	0	1	-3	2	-10
X_2	2	0	-1	1	0	2	-1	10

1

Bas Var	Eq No	Coefficient of						Right side
		Z	X_1	X_2	X_3	X_4	X_5	
Z	0	1	-2	-7	-4	0	0	0
X_4	1	0	1	2	1	1	0	10
X_5	2	0	3	3*	2	0	1	10

2

Bas Var	Eq No	Coefficient of						Right side
		Z	X_1	X_2	X_3	X_4	X_5	
Z	0	1	5	0	0.667	0	2.333	23.33
X_4	1	0	-1	0	-0.33*	1	-0.67	3.333
X_2	2	0	1	1	0.667	0	0.333	3.333

Complementary Basic Solution - Properties

If one variable is basic, its associated variable in the complementary BS is non-basic, therefore zero. Why? Let x_j be a BV in a primal simplex tableau. Then its coefficient in eq.(0) of that tableau is zero which is the value of the i th surplus variable y_{m+j} (which is associated with x_j).

If slack x_{n+i} is a BV, its coefficient in eq.(0) is zero, which is the value of the associated dual variable y_i in the complementary dual BS.

Complementary Basic Solution – Properties Cont.

- Complementary basic solutions property: A primal BS is associated with a complementary BS in the dual problem, where the respective objective function values (z and y_0) are equal
- Complementary slackness property: Given two complementary BS (primal and dual) in each pair of associated variables (one primal and one dual), one is basic and the other is non-basic, or:

$$\text{if } x_j \neq 0 \Rightarrow y_{m+j} = 0; \text{ if } x_{n+i} \neq 0 \Rightarrow y_i = 0; \text{ or } x_j * y_{m+j} = 0 \text{ and } x_{n+i} * y_i = 0, \\ \forall i, j$$

Iteration	Basic Variable	Eq. No.	z	Coefficient of					Right Side
				x_1	x_2	x_3	x_4	x_5	
0	z	0	1	-3	-5	0	0	0	0
	x_3	1	0	1	0	1	0	0	4
	x_4	2	0	0	2	0	1	0	12
	x_5	3	0	3	2	0	0	1	18
1				y_4	y_5	y_1	y_2	y_3	y_b
	z	0	1	-3	0	0	5/2	0	30
	x_3	1	0	1	0	1	0	0	4
	x_2	2	0	0	1	0	1/2	0	6
	x_5	3	0	3	0	0	-1	1	6
2				y_4	y_5	y_1	y_2	y_3	y_b
	z	0	1	0	0	0	3/2	1	36
	x_3	1	0	0	0	1	1/3	-1/3	2
	x_2	2	0	0	1	0	1/2	0	6
	x_1	3	0	1	0	0	-1/3	1/3	2

Complementary Basic Solutions

Step 1: Assign to the dual variables of the Prototype problem, the coefficients of the associated primal variables (slacks) in the (0) row of It.(1): $y_1=0$, $y_2=5/2$, $y_3=0$, or $y=[0, 5/2, 0]$. The coefficients of the primal variables x_1 and x_2 in the (0) row of the tableau are the values of the surplus variables of the dual (y_4, y_5).

$$s = [y_4, y_5] = yA - c = \begin{bmatrix} 0 & 5/2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 5 \end{bmatrix} = \begin{bmatrix} -3 & 0 \end{bmatrix}$$

Complementary Basic Solutions Cont.

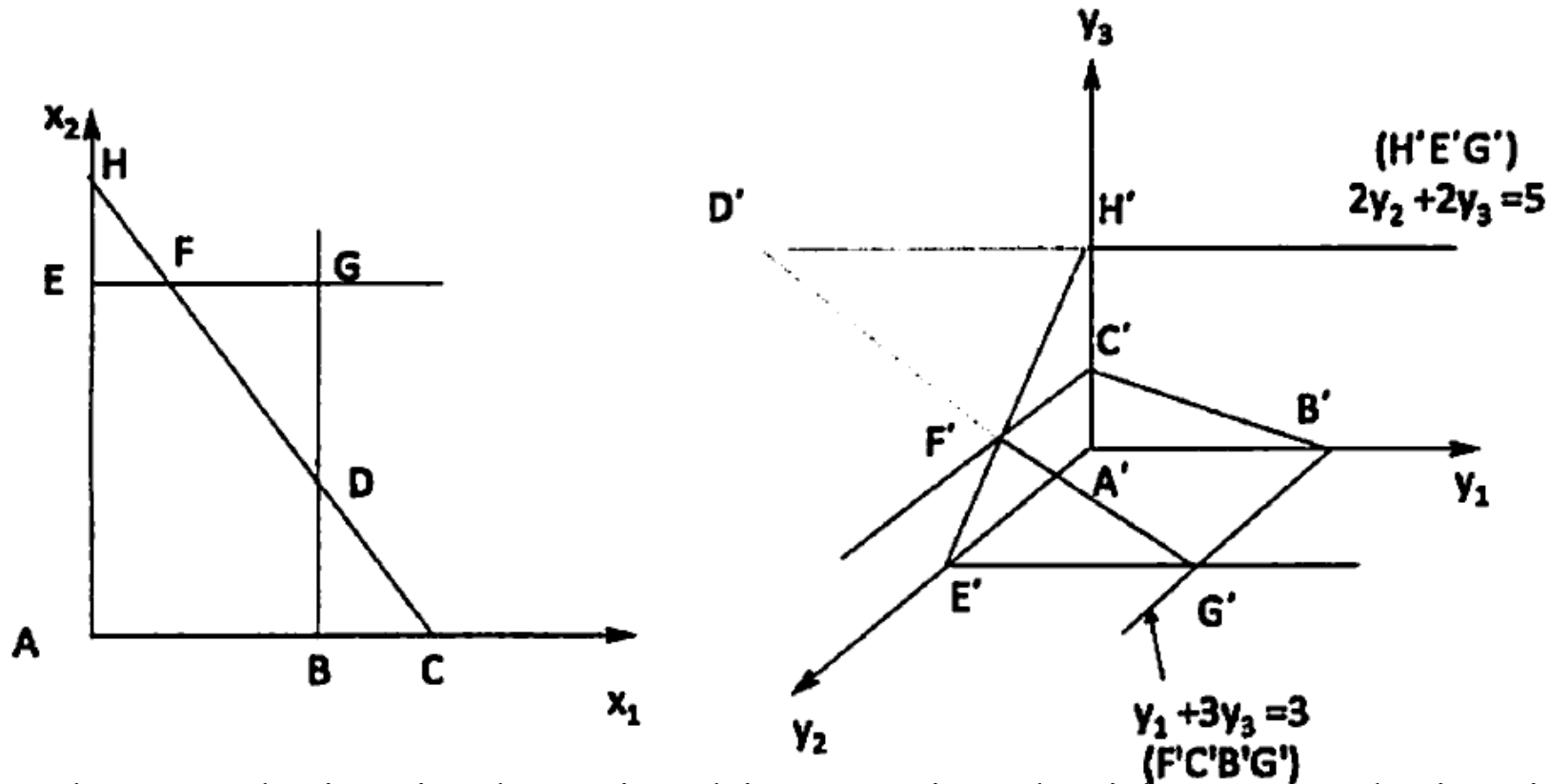
Step 2: The objective values of the two problems (primal and dual) for these complementary basic solutions are equal.

$$y_0 = yb = \begin{bmatrix} 0, & 5/2, & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} = 30 = cx = \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = z$$

Step 3: Complementary at in each column, we have associated primal-dual variables and characterize them as BV or NB variables.

NB	BV	BV	NB	BV	
$x_1=0,$	$x_2=6$	$x_3=4,$	$x_4=0,$	$x_5=6$	$z=30$
$y_4=-3,$	$y_5=0$	$y_1=0,$	$y_2=5/2,$	$y_3=0$	$y_0=30$
BV	NB	NB	BV	NB	

Graphical Solution of Primal & Dual



Each CP solution in the primal is associated with a CP solution in the dual: $A \rightarrow A'$, $B \rightarrow B'$, ... , $H \rightarrow H'$ and the corresponding primal-dual BS are complementary.

Generating Complementary Basic Solutions

We can generate all complementary primal-dual BS systematically by:

Setting 2 out of the 5 primal variables to zero (x_1, x_2, x_3, x_4, x_5), by making them NB and solving the system of the 3 primal equations in the remaining 3 variables,

Finding the complementary dual BS, using complementary slackness.

Example: Set $x_3 = x_5 = 0$ and substitute in the primal equations: $x_1 + 0 = 4$, $2x_2 + x_4 = 12$, $3x_1 + 2x_2 + 0 = 18$, which provides the remaining variables $x_1 = 4$, $x_2 = 3$, $x_4 = 6$.

Generating Complementary Basic Solutions Cont.

The resulting primal solution $(x_1, x_2, x_3, x_4, x_5) = (4, 3, 0, 6, 0)$ is feasible and is associated with CP solution D.

To find the complementary dual BS, we construct two rows for the two BSs, and align vertically in pairs the associated primal-dual variables and write the values of the primal variables as well as their status.

Using complementary slackness, we determine the status of the dual variables and set the NB variables to zero:

BV		NB		BV		NB	
$x_1=4,$	$x_2=3$	$x_3=0,$	$x_4=6,$	$x_5=0$			
$y_4=0,$	$y_5=0$	$y_1=?,$	$y_2=0,$	$y_3=?$			

Generating Complementary Basic Solutions Cont.

The remaining two variables are determined by substituting the zeros to the dual equations and solving: $y_1 + 3y_3 - 0 = 3$, and $2(0) + 2y_3 - 0 = 5$,

which yield $y_1 = -9/2$ and $y_3 = 5/2$. The dual complementary solution is infeasible and is associated with D' .

The two objective values should be equal; $z = (3)(4) + (5)(3) = 27$ & $y_0 = (4)(-9/2) + (12)(0) + (18)(5/2) = 27$.

List of Complementary Basic Solutions

Primal-dual complementary BSs - prototype example

No	Pt	$(x_1, x_2, x_3, x_4, x_5)$	Feasible?	$z=y_0$	Feasible?	$(y_4, y_5, y_1, y_2, y_3)$	Pt
8	H	(0, 9, 4, -6, 0)	N	45	Y	(9/2, 0, 0, 0, 5/2)	H'
7	G	(4, 6, 0, 0, -6)	N	42	Y	(0, 0, 3, 5/2, 0)	G'
6	F	(2, 6, 2, 0, 0)	Y	36	Y	(0, 0, 0, 3/2, 1)	F'
5	E	(0, 6, 4, 0, 6)	Y	30	N	(-3, 0, 0, 5/2, 0)	E'
4	D	(4, 3, 0, 6, 0)	Y	27	N	(0, 0, -9/2, 0, 5/2)	D'
3	C	(6, 0, -2, 12, 0)	N	18	N	(0, -3, 0, 0, 1)	C'
2	B	(4, 0, 0, 12, 6)	Y	12	N	(0, -5, 3, 0, 0)	B'
1	A	(0, 0, 4, 12, 18)	Y	0	N	(-3, -5, 0, 0, 0)	A'
Primal					Dual		

Classification of Basic Solutions on $z=y_0$

PRIMAL			DUAL	
Infeasible superoptimal			Suboptimal	
NF	(H) 45	- 45 (H')	F	
NF	(G) 42	- 42 (G')	F	
F	(F) 36	- 36 (F')	F	optimum
optimum	F	(E) 30 - 30 (E')	NF	
	F	(D) 27 - 27 (D')	NF	
Suboptimal			Infeasible superoptimal	
NF	(C) 18	- 18 (C')	NF	
F	(B) 12	- 12 (B')	NF	
F	(A) 0	- 0 (A')	NF	

Classification of Basic Solutions Cont.

Primal infeasible superoptimal quadrant (northwest) contains only infeasible BSs.

Southeast space contains only infeasible dual BSs.

Why? Because the objective value is better than 36 (higher for the primal and lower for the dual). If any one of these solutions was feasible, it would have contradicted with the primal/dual objective value of 36.

Classification of Basic Solutions Cont.

Only infeasible BSs should appear in those two quadrants, for the primal and dual problem.

This provided the motivation for the discovery of the dual simplex method.

However, the other two quadrants may contain both feasible and infeasible BSs.

Complementary Optimal Solutions

Each simplex tableau is associated with a BS for the primal problem and a complementary BS for the dual problem.

As we move from one iteration to the next, z improves (increases) while the dual BS improves in feasibility, i.e., it has fewer negative values for the dual variables. When there are no negative values in the dual variables (coefficient in eq.(0)), we reached optimality for the primal while we attained feasibility for the dual problem.

Iteration	Basic Variable	Eq. No.	z	Coefficient of					Right Side
				x_1	x_2	x_3	x_4	x_5	
0	z	0	1	-3	-5	0	0	0	0
	x_3	1	0	1	0	1	0	0	4
	x_4	2	0	0	2	0	1	0	12
	x_5	3	0	3	2	0	0	1	18
1				y_4	y_5	y_1	y_2	y_3	y_b
	z	0	1	-3	0	0	5/2	0	30
	x_3	1	0	1	0	1	0	0	4
	x_2	2	0	0	1	0	1/2	0	6
	x_5	3	0	3	0	0	-1	1	6
2				y_4	y_5	y_1	y_2	y_3	y_b
	z	0	1	0	0	0	3/2	1	36
	x_3	1	0	0	0	1	1/3	-1/3	2
	x_2	2	0	0	1	0	1/2	0	6
	x_1	3	0	1	0	0	-1/3	1/3	2

Complementary Optimal Solutions Cont.

Weak Duality Theorem: $cx \leq yb$ for any two feasible solutions (x and y) of the primal and dual problem. Thus, the y vector in the optimal primal simplex tableau should be the optimal solution of the dual problem, since yb cannot be lower than cx^* and we just found a feasible y such that $yb = cx^*$.

At the final iteration, Simplex finds both an optimal solution x^* for the primal problem, and a complementary optimal solution y^* for the dual problem (coefficients of slack var. in row 0), where $cx^* = y^*b$, and y^* is the vector of the shadow prices of the primal problem.

Additional Primal-Dual Properties

Strong duality property: If x^* is an optimal solution for the primal problem and y^* is an optimal solution for the dual problem, then $z = cx^* = y^*b = y_0$.

Symmetry property: For any primal problem and its dual, all relationships between them are symmetric because the dual of the dual is the primal problem

Infeasible/unbounded problem: If one of the two problems is unbounded, the other is infeasible. If one of the two problems is infeasible, the other is either unbounded or infeasible.

Karush-Kuhn-Tucker (KKT) Conditions

A solution for the primal $[x, xs]$ and a solution for the dual $[y, s]$ are optimal for the respective problems if they satisfy the following 3 KKT conditions:

- a. Primal feasibility: $(Ax + xs = b; x, xs \geq 0)$
- b. Dual feasibility: $(yA - s = c; y, s \geq 0)$
- c. Complementary slackness: $(yxs + sx = 0)$

KKT Conditions & Simplex Algorithms

All simplex algorithms are based on the KKT conditions.

Regular simplex algorithm: We maintain primal feasibility & complementary slackness, while we are striving for dual feasibility.

Dual simplex algorithm: We maintain dual feasibility and complementary slackness and strive for primal feasibility.

Finally, it is possible to maintain primal and dual feasibility and strive for complementary slackness. This is the third simplex algorithm, known as primal-dual simplex algorithm.

KKT Conditions: Prototype Example

!Primal Feasibility;

$$x_1 + x_3 = 4;$$

$$2x_2 + x_4 = 12;$$

$$3x_1 + 2x_2 + x_5 = 18;$$

!Dual Feasibility;

$$y_1 + 3y_3 - y_4 = 3;$$

$$2y_2 + 2y_3 - y_5 = 5;$$

!Complementary Slackness;

$$y_1x_3 + y_2x_4 + y_3x_5 + y_4x_1 + y_5x_2 = 0;$$

Lagrangian Multipliers

Duality theory is generalized for nonlinear programming

Dual variables are named generalized dual variables or Lagrangian multipliers

KKT conditions for nonlinear programming provide the foundation for the development of nonlinear programming algorithms

Adapting to Other Primal Forms

Dual transformation of a "=" constraint

(P)

$$\begin{aligned} \text{Max } z &= 3x_1 + 5x_2 \\ \text{s.t. } x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &= 18 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$\equiv$

$$\text{Max } z = 3x_1 + 5x_2$$

$$\begin{aligned} \text{s.t. } x &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \\ -3x_1 - 2x_2 &\leq -18 \\ x_1, x_2 &\geq 0 \end{aligned}$$



Dual (P)

$$\begin{aligned} \text{Min } y_0 &= 4y_1 + 12y_2 + 18y_3 \\ \text{s.t. } y_1 &+ 3y_3 \geq 3 \\ 2y_2 + 2y_3 &\geq 5 \\ y_1, y_2 &\geq 0 \\ y_3 &\text{ unconstrained in sign} \end{aligned}$$

$\equiv$

Dual (P)

$$\begin{aligned} \text{Min } y_0 &= 4y_1 + 12y_2 + 18y_4 - 18y_5 \\ \text{s.t. } y_1 &+ 3y_4 - 3y_5 \geq 3 \\ 2y_2 + 2y_4 - 2y_5 &\geq 5 \\ y_1, y_2, y_4, y_5 &\geq 0 \end{aligned}$$

Transformation:

$$y_3 = y_4 - y_5$$

y_3 unconstrained in sign

Dual Transformation: "=" Constraint

- Dual Symmetric Relationships
- The dual variable associated with an "=" primal constraint is unconstrained in sign
- Since the $\text{Dual}[\text{Dual}(P)] = P$, the dual constraint associated with an unconstrained in sign primal variable is "=" constraint.

Adapting to Other Primal Forms Cont.

Dual transformation of a “ $\geq$ ” constraint in a “max” LP

$$\begin{array}{ll}
 \text{(P)} & \\
 \text{Max } z = & 3x_1 + 5x_2 \\
 \text{s.t.} & x_1 \leq 4 \\
 & 2x_2 \leq 12 \\
 & 3x_1 + 2x_2 \geq 18 \\
 & x_1, x_2 \geq 0
 \end{array}$$

 $\equiv$

$$\begin{array}{ll}
 \text{(P)} & \\
 \text{Max } z = & 3x_1 + 5x_2 \\
 \text{s.t.} & x_1 \leq 4 \\
 & 2x_2 \leq 12 \\
 & -3x_1 - 2x_2 \leq -18 \\
 & x_1, x_2 \geq 0
 \end{array}$$



$$\begin{array}{ll}
 \text{Dual (P)} & \\
 \text{Min } y_0 = & 4y_1 + 12y_2 + 18y_3 \\
 \text{s.t.} & y_1 + 3y_3 \geq 3 \\
 & 2y_2 + 2y_3 \geq 5 \\
 & y_1, y_2 \geq 0 \\
 & y_3 \leq 0
 \end{array}$$

 $\equiv$

$$\begin{array}{ll}
 \text{Dual (P)} & \\
 \text{Min } y_0 = & 4y_1 + 12y_2 - 18y_4 \\
 \text{s.t.} & y_1 - 3y_4 \geq 3 \\
 & 2y_2 - 2y_4 \geq 5 \\
 & y_1, y_2, y_4 \geq 0
 \end{array}$$

$$y_3 = -y_4 \leq 0$$

Dual Transformation: " $\geq$ " Constraint

- Dual Symmetric Relationships
- The dual variable associated with a " $\geq$ " constraint in a max primal problem is " ≤ 0 "
- Since the $\text{Dual}[\text{Dual}(P)] = P$, the dual constraint associated with a " ≤ 0 " primal variable is incompatible to the dual objective (If the dual objective is "max", the dual constraint is " $\geq$ "; if the dual objective is "min", the dual constraint is " $\leq$ ").

Dual Transformations

Primal (Dual)	Dual (Primal)
max z	min y_0
Constraint i : $\leq$ form $=$ form $\geq$ form	Variable y_i : $y_i \geq 0$ unconstrained y_i $y_i \leq 0$
Variable x_j : $x_j \geq 0$ unconstrained x_j $x_j \leq 0$	Constraint j : $\geq$ form $=$ form $\leq$ form

Dual Transformation Example

Primal (P)

$$\text{Min } W = x_1 + 2x_2 + 3x_3$$

$$\text{s.t.} \quad x_2 + 3x_3 = 5$$

$$2x_1 + x_2 - x_3 \leq 10$$

$$-x_1 - x_2 + 2x_3 \geq 0$$

$$x_1 \geq 0$$

$$x_2 \leq 0$$

$$x_3 \leq 0 \text{ unrestricted in sign}$$

Dual Simplex Algorithm

PRIMAL				DUAL			
Infeasible superoptimal				Suboptimal			
NF	(H)	45	45 (H')	F			
NF	(G)	42	42 (G')	F			
F	(F)	36	36 (F')	F		optimum	
F	(E)	30	30 (E')	NF			
F	(D)	27	27 (D')	NF			
NF	(C)	18	18 (C')	NF			
F	(B)	12	12 (B')	NF			Infeasible superoptimal
F	(A)	0	0 (A')	NF			
Suboptimal				optimum			

Dual of the Prototype Example

$$\begin{aligned}
 \min z &= 4y_1 + 12y_2 + 18y_3 \\
 \text{s.t.} \quad &y_1 + 3y_3 \geq 3 \\
 &2y_2 + 2y_3 \geq 5 \\
 &y_i \geq 0, i = 1, 2, 3
 \end{aligned}$$

$$\begin{aligned}
 \text{or max } -z &= -4y_1 - 12y_2 - 18y_3 \\
 \text{s.t.} \quad &y_1 + 3y_3 - y_4 = 3 \\
 &2y_2 + 2y_3 - y_5 = 5 \\
 &y_i \geq 0, i = 1, \dots, 5
 \end{aligned}$$

$$\begin{aligned}
 \text{or max } -z \\
 \text{s.t. } -z + 4y_1 + 12y_2 + 18y_3 &= 0 \\
 -y_1 - 3y_3 + y_4 &= -3 \\
 -2y_2 - 2y_3 + y_5 &= -5 \\
 y_i &\geq 0, i = 1, \dots, 5
 \end{aligned}$$

Dual Simplex Algorithm Cont.

It starts with an infeasible but superoptimal BS. In successive iterations we move to a BS that improves on feasibility while maintaining superoptimality.

The first feasible BS found is the optimal.

Iter.#	BV	Eq. #	z	Coefficient of					RHS
				y_1	y_2	y_3	y_4	y_5	
0	z	0	-1	4	12	18	0	0	0
	y_4	1	0	-1	0	-3	1	0	-3
	y_5	2	0	0	-2*	-2	0	1	<u>-5</u>

most negative is -5 for pivot row (2); $\min(|12/(-2)|, |18/(-2)|) = 6$ for y_2 .

1	z	0	-1	4	0	6	0	6	-30
	y_4	1	0	-1	0	-3*	1	0	<u>-3</u>
	y_2	2	0	0	1	1	0	-1/2	5/2

2	z	0	-1	2	0	0	2	6	-36
	y_3	1	0	1/3	0	1	-1/3	0	1
	y_2	2	0	-1/3	1	0	1/3	-1/2	3/2

No negative RHS, $\Rightarrow$ STOP.

Optimal solution: $(y_1, y_2, y_3, y_4, y_5, z) = (0, 3/2, 1, 0, 0, 36)$.

Primal Simplex:

Iteration	Basic Variable	Eq. No.	z	Coefficient of					Right Side
				x_1	x_2	x_3	x_4	x_5	
2				y_4	y_5	y_1	y_2	y_3	y_b
	z	0	1	0	0	0	3/2	1	36
	x_3	1	0	0	0	1	1/3	-1/3	2
	x_2	2	0	0	1	0	1/2	0	6
	x_1	3	0	1	0	0	-1/3	1/3	2

Dual Simplex:

Iter.#	BV	Eq. #	z	Coefficient of					RHS
				y_1	y_2	y_3	y_4	y_5	
2	z	0	-1	2	0	0	2	6	-36
	y_3	1	0	1/3	0	1	-1/3	0	1
	y_2	2	0	-1/3	1	0	1/3	-1/2	3/2

$x_1 \leftrightarrow y_4$
 $x_2 \leftrightarrow y_5$
 $x_3 \leftrightarrow y_1$
 $x_4 \leftrightarrow y_2$
 $x_5 \leftrightarrow y_3$

Optimal Solution from Dual Simplex

What is the information contained in row(0)?

Compare the operations performed using Dual Simplex on the dual of the problem with the regular Simplex operations performed on the primal of the problem .

Iteration 0 of Primal Simplex:

Iteration	Basic Variable	Eq. No.	z	Coefficient of					Right Side
				x_1	x_2	x_3	x_4	x_5	
0	z	0	1	-3	-5	0	0	0	0
	x_3	1	0	0	0	1	0	0	4
	x_4	2	0	0	2	0	1	0	12
									18

$x_1 \leftrightarrow y_4$
 $x_2 \leftrightarrow y_5$
 $x_3 \leftrightarrow y_1$
 $x_4 \leftrightarrow y_2$
 $x_5 \leftrightarrow y_3$

Iteration 0 of Dual Simplex:

Iter.#	BV	Eq. #	z	Coefficient of					RHS
				y_1	y_2	y_3	y_4	y_5	
0	z	0	-1	4	12	18	0	0	0
	y_4	1	0	-1	0	-3	1	0	-3
	y_5	2	0	0	-2*	-2	0	1	-5

Dual Simplex Algorithm Cont.

What if there is no negative element in the pivot row?

1	z	0	-1	4	0	6	0	6	-30
	y_4	1	0	+1	0	+3*	1	0	<u>-3</u>
	y_2	2	0	0	1	1	0	-1/2	5/2

Dual Simplex Algorithm Cont.

Do NOT transform the problem to its dual to use the Dual Simplex Method. It is a method by itself.

To Use the Dual Simplex Method:

LP	Objective Function Coefficients
Min	Positive or Zero
Max	Negative or Zero

Work on Module 6 HW 6

Module 7 Reading: Section 6.4, 7.1, 7.2 & 7.7 to learn about sensitivity analysis.